

Transrex Upgrade Plan

1 Introduction

The Transrex system consists of four Transrex ISR-2126 magnet power supplies connected in parallel which power the two Phaedrus mirror coils and the two W7A central trim coils.

2 Existing System

The original analog control systems in the four ISR-2126 power supplies at WHAM were replaced with modern 12-pulse SCR firing boards manufactured by Enerpro. This firing board accepts a 0-10 V analog signal which corresponds to phase angles from 180 to 0 degrees. This 0-10 V signal comes from a custom 'error amplifier' board, which serves as an interface board between the main Transrex control system, the Enerpro firing board, and the two LEMs that monitor currents between the two bridges.

The existing system operates completely open-loop. An overcurrent on any one of the four Transrexes will immediately shut down all four units, and any line voltage variations will have a proportional effect on the output current.

3 Problems

There are a number of problems with the existing system that necessitate a substantial system upgrade.

- The existing open-loop control makes the current through the central field coils less repeatable.
- The existing fault handling system, which immediately shuts down all four units as soon as any individual unit detects an issue leads to very fast ramp-down which puts undue stress on the HTS magnets.
- The existing controls are vulnerable to EMI, leading to false trips of the overcurrent protection (OCP) when other systems in the lab experience a malfunction.
- The existing current measurements are inaccurate and incomplete.
- The interlocks for SCR temperature and water flow are not operational.
- There is a substantial imbalance between the two bridge currents in one of the four units (Laa-Laa), leading to increased 360 Hz ripple.
- There are a number of faulty SCRs in at least one of the four units, leading to increased subharmonic ripple.
- Operation of the Transrexes uses a large amount of I/O on the main CRIO control system.

4 Key Requirements

- Temperature and water-cooling interlocks must be functional and shut system down in a safe, controlled manner.
- Functional emergency stop (E-stop).
- Closed-loop control.
- If an individual unit senses an overcurrent fault, the other three units should begin a graceful ramp-down rather than immediately shutting off.
- Interlocks and overcurrent protection must be resistant to RF interference from ECH, NBI and Archimedes systems.

5 Upgrades

- 5kA output LEMs on each Transrex
- New interface boards
- New control unit
- New digitizer units in each Transrex, rather than a single large digitizer.
- Main 20 kA LEM on transmission line for global system feedback
- All connections between Transrexes and control unit are over fiber optic to avoid ground loops.

6 Interface Boards

The interface boards connect to the two bridge LEMs, the 5 kA output LEM, the firing board, the two interlock strings, as well as to the control unit. A diagram of the signal flow for the interface board can be seen in Fig. 1

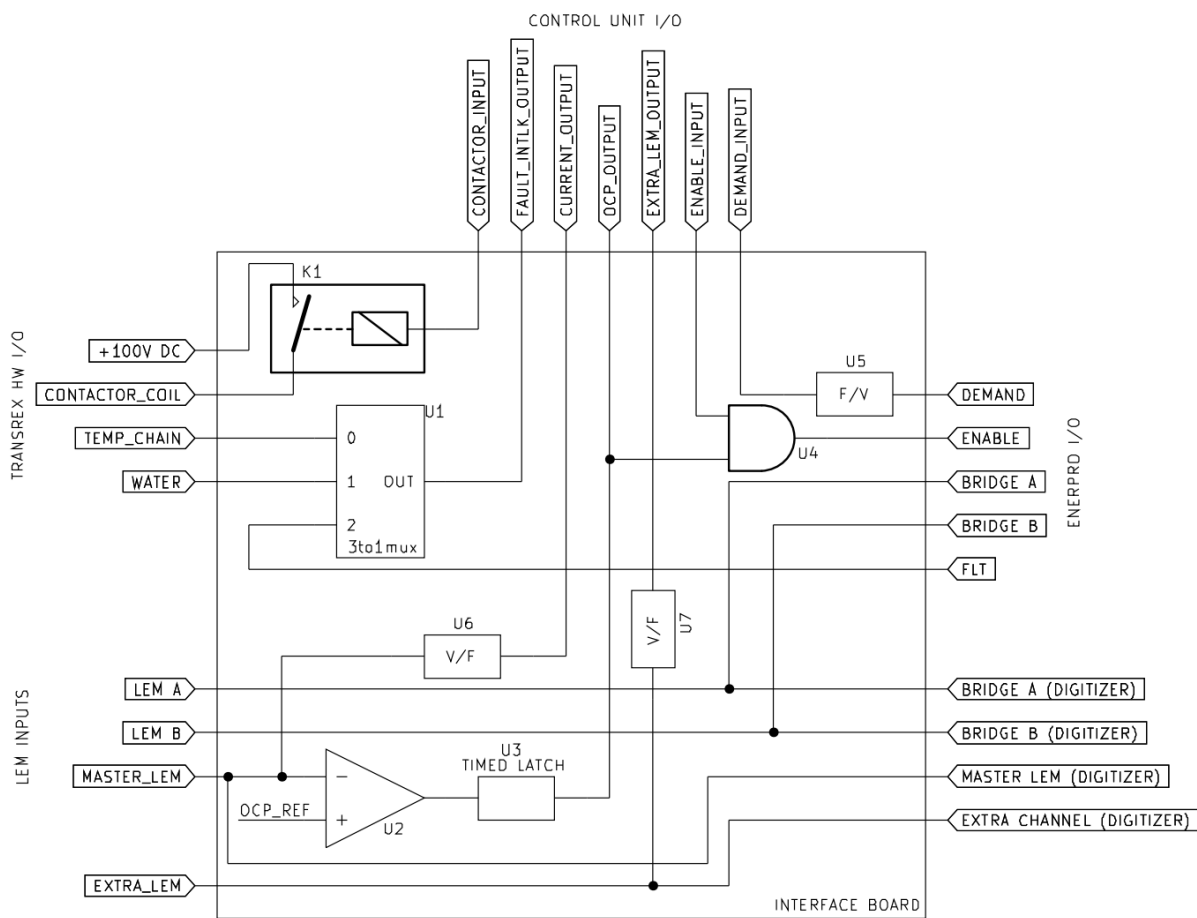


Fig. 1. Shows the basic signal routing for the interface board.

6.1 Fault Handling

The overcurrent protection comparator directly interlocks the Enerpro enable signal, which locks out for 120 seconds when an overcurrent is detected.

The interlocks for temperature switches and water flow are fed back to the control unit, which responds accordingly to an open interlock chain.

6.2 Interface Board I/O

Signal Name	Direction	Format	Description
MASTER_LEM	IN	Analog, 0-5 V	Input connection for 5 kA LEM
EXTRA_LEM	IN	Analog, 0-5 V	Input connection for 20 kA LEM or voltage LEM
LEM_A	IN	Analog, 0-5 V	Input connection for Bridge A LEM
LEM_B	IN	Analog, 0-5 V	Input connection for Bridge B LEM
TEMP	IN	RELAY	Temperature switch chain input connection
H ₂ O	IN	RELAY	Water flow switch input connection
FLT	IN	TTL	Enerpro fault indicator.
CONTACTOR	OUT	RELAY	Relay for controlling main contactor.
DEMAND	OUT	Analog, 0-10 V	Firing angle control output to Enerpro
ENABLE	OUT	Digital, 0 / 10 V	Enable signal for Enerpro.
BRIDGE_A	OUT	Analog, 0-10 V	Analog bridge reference signal for Enerpro
BRIDGE_B	OUT	Analog, 0-10 V	Analog bridge reference signal for Enerpro
CURRENT_OUT	OUT	Fiber Optic, 0 - 100 kHz	Supply output current to control unit
OCP_OUT	OUT	Digital fiber optic	Overcurrent protection output indicator to control unit
FLT_INTLK	OUT	Fiber Optic, Frequency	Multiplexed H ₂ O, Temperature and Enerpro faults to control unit.
ENABLE_IN	IN	digital fiber optic	Enerpro enable from control unit
CONTACTOR_IN	IN	digital fiber optic	Contactors control from control unit
DEMAND_IN	IN	Fiber optic, 0-100 kHz	Demand signal from the control unit which controls the Enerpro.
BRIDGE A LEM (DIGITIZER)	OUT	Analog, 0-10 V	Output to digitizer.
BRIDGE B LEM (DIGITIZER)	OUT	Analog, 0-10 V	Output to digitizer.
MASTER LEM (DIGITIZER)	OUT	Analog, 0-10 V	Output to digitizer.
EXTRA CHANNEL (DIGITIZER)	OUT	Analog, 0-10 V	Output to digitizer.

7 Control Unit

The control unit receives commands from the main LabVIEW control system and interfaces with the four interface boards to control the Transrex output currents. All of the control unit I/O is done over fiber optic connections.

7.1 Control Unit I/O

Signal Name	Direction	Format	Quantity	Description
UNIT_CURRENT	IN	Fiber Optic, 0 - 100 kHz	4	Output current for each Transrex supply.
FAULT_INTLK	IN	Fiber Optic, Frequency	4	Multiplexed H ₂ O, Temperature and Enerpro faults for each Transrex.
CONTACTOR	OUT	Digital fiber optic	4	Contactor control for each Transrex.
OCP	IN	Digital fiber optic	4	OCP status from each Transrex.
ENABLE	OUT	Digital fiber optic	4	Enerpro enable for each Transrex.
DEMAND	OUT	Fiber Optic, 0 - 100 kHz	4	Firing angle for each Transrex.
SYS_ENABLE	IN	Digital fiber optic	1	Global system enable, from CRIO control system.
SHOT_START	IN	Digital fiber optic	1	System 'go' signal from CRIO control system.
E-STOP	IN	Digital fiber optic	1	Emergency stop signal from manual E-stop switches.
COMMS	IN/OUT	Fiber optic Ethernet	1	Communication with LabVIEW

7.2 Feedback Loops

The DEMAND_IN signal is proportional to the firing angle. The total system output is measured by a 20 kA LEM, to be installed at the corner of the transmission line, and a digital PID loop is implemented in the control unit to adjust the four DEMAND_IN signals accordingly. In the event that one of the four units detects an overcurrent, the control unit must decrease the DEMAND_IN signals by 25% so that the three remaining units do not see an increase in output current.

7.3 Communications Protocol

There are two types of signal encoding over versatile link fiber optic that will be utilized. The DEMAND and I_OUT signals utilize voltage-to-frequency conversion, allowing analog signals to be transmitted over fiber optic, with a frequency range from 100 Hz to 100 kHz, with 100 kHz representing full scale and 100 Hz representing zero.

In order to limit the number of fiber optic connections required for low-speed signal information, three separate fault interlocks (water, temperature and the Enerpro status) are all carried on a single fiber optic line using Frequency Division Multiplexing (FDM). The water interlock is encoded at 500 Hz, temperature at 5 kHz, and Enerpro status at 50 kHz.

7.4 Fault Handling

It is critical that the E-stop functionality be completely independent of software, such that a microprocessor crash or lockup does not inhibit the ability for an operator to stop the system in an emergency.

Fault	Signal Line	Response
Overcurrent	OCP_OUT	Immediately reduce DEMAND_IN signal to all other Transrex units by 25%. Initiate ramp down.
Overcurrent (2+ units)	OCP_OUT	Immediately reduce DEMAND_IN signal to all other Transrex units by 25% times the number of faulted units. Initiate ramp down.
Enerpro fault	FLT_INTLK	Immediately reduce DEMAND_IN signal to all other Transrex units by 25%. Initiate ramp down.
Enerpro fault (2+ units)	FLT_INTLK	Immediately reduce DEMAND_IN signal to all other Transrex units by 25% times the number of faulted units. Initiate ramp down.
Water and/or temperature fault	FLT_INTLK	Immediately initiate ramp-down signal to all four units.
E-stop Pressed	N/A	ENABLE_IN and CONTACTOR_IN fibers are immediately set low.

8 Digitizers

The original Transrex system utilized a 16-channel D-Tacq located in the main control unit, with all of the digitized signals being sent over differential pairs between each Transrex and the control unit. While functional, this has not proven to be as electrically robust as is desired.

As part of the upgrade, the single, 16-channel digitizer in the control unit will be replaced with a four-channel digitizer in each Transrex. Siglent SDS800X series oscilloscopes have successfully been used as digitizers for other WHAM systems, and have sufficient record length to digitize the three required signals (LEM A, LEM B, MASTER_LEM) with good resolution for the 20-30 second shot cycle. These oscilloscopes will be powered from the internal 120 V power inside each Transrex, with the scopes at local (Transrex) ground potential. They will communicate with the WHAM data system over fiber optic ethernet. Rack mounting kits (part number SDS1000X-E-RMK) allow these scopes to be securely mounted in place of the original NIM control system, which is currently covered by a blank panel.

The 20 kA LEM and the transmission line voltage measurement are each powered and digitized by one of the four Transrex units, making use of the extra digitizer channel that each unit has.

9 Other Work

In addition to the control system upgrades outlined above, the following upgrades and maintenance procedures are to be carried out during the project.

- Verify all SCRs are firing as intended, and replace any faulty SCRs.
- Replace DC control power supplies.
- Verify that the bridge LEMs in all units are identical.
- Test all temperature switches, and replace any that are open to restore temperature interlock functionality.
- Calibrate flow switches.

10 System Block Diagram

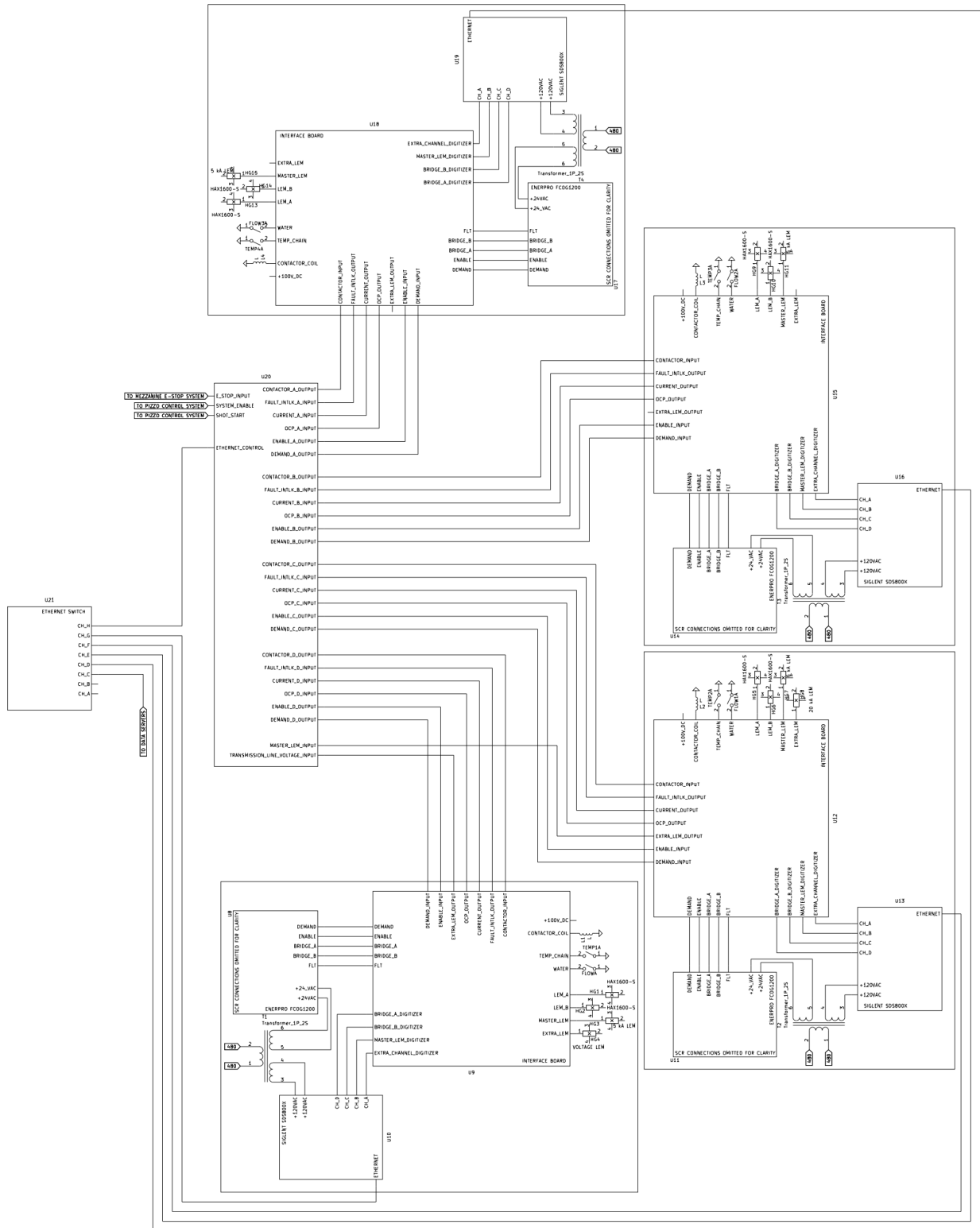


Fig. 2. Shows the architecture of the Transrex system.

11 Revision History

Revision	Name	Date	Modification
1.0	Oscar L Anderson	2026-07-10	Creation